

Subject-Agnostic fMRI Visual Reconstruction via Shared Latent Alignment and CLIP–DINOv2 Guided Diffusion

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Abstract

Reconstructing visual information from functional magnetic resonance imaging (fMRI) remains challenging because neural responses are high-dimensional and vary substantially across subjects. Existing brain-to-image methods often rely on subject-specific decoding, limiting their ability to generalize across individuals and datasets. This work presents a subject-agnostic multi-subject fMRI visual reconstruction framework that integrates shared latent alignment, Transformer-based neural encoding, CLIP–DINOv2 semantic–structural feature prediction, dual diffusion-prior refinement, multimodal feature fusion, retrieval-guided matching, and Stable Diffusion-based image reconstruction. The framework learns a common neural representation to reduce inter-subject variability and combines complementary semantic and structural visual features to guide reconstruction. Evaluation was conducted on three complementary fMRI datasets: the Haxby dataset for category-level visual decoding, the Natural Scenes Dataset (NSD) for large-scale natural-scene reconstruction, and a NIMHANS-acquired LIVE fMRI dataset for independent subject-wise and cross-dataset evaluation. The proposed framework achieved average SSIM/PixCorr values of 0.8184/0.6086 on Haxby, 0.5333/0.4304 on NSD, and 0.3917/0.3707 on LIVE, respectively. On LIVE, the framework additionally obtained 77.56% and 82.60% similarity at AlexNet layers 2 and 5, and 74.16% CLIP similarity. NSD data-availability experiments further showed progressive improvements across the 1 h, 7 h, and 40 h settings. Overall, the results demonstrate that combining shared latent alignment with semantic–structural feature fusion and retrieval-guided diffusion provides a unified framework for preserving visual content and supporting cross-subject and cross-dataset fMRI visual reconstruction.

Keywords: fMRI Decoding; Brain-to-Image Reconstruction; Multi-Subject Learning; Transformer Networks; CLIP; DINOv2; Retrieval-Guided Reconstruction; Diffusion Models; Stable Diffusion; Computational Neuroscience.

1 Introduction

The process by which the brain interprets visual information remains an active area of scientific research. Functional magnetic resonance imaging (fMRI) allows brain activity to be studied without surgery by recording changes in blood-oxygen-level-dependent (BOLD) signals that occur during visual perception [1,2]. Examination of these signals has revealed that different visual experiences produce distinguishable activation patterns, enabling computational models to identify the content of viewed images from brain responses [3–5]. In addition, evidence suggests that the visual cortex encodes

both perceptual and semantic properties of visual information, providing a strong foundation for advances in brain decoding and visual reconstruction [6].

Initial fMRI decoding methods primarily relied on conventional machine learning techniques, including support vector machines, Bayesian inference, and dimensionality reduction, to establish mappings between neural responses and visual representations [7, 28, 29]. Although these approaches demonstrated the possibility of decoding visual information, their limited ability to model complex nonlinear relationships often resulted in low reconstruction fidelity and weak semantic represen-

tation.

The introduction of deep learning significantly improved fMRI decoding by enabling hierarchical feature learning from high-dimensional neural signals. Convolutional neural networks and deep neural architectures achieved more accurate mappings between brain activity and visual features, leading to substantial improvements in reconstruction quality and semantic consistency [9–13]. More recently, multimodal foundation models have further advanced brain-to-image reconstruction. CLIP provides robust semantic visual representations through large-scale image-text contrastive learning, while DINO and DINOv2 learn rich structural features using self-supervised vision transformers, allowing complementary modeling of semantic concepts and fine-grained visual structures [14, 17, 18]. Combining these complementary representations has shown considerable potential for improving reconstruction quality.

In parallel, diffusion-based generative models have become the dominant paradigm for high-quality image synthesis. Denoising diffusion probabilistic models and latent diffusion frameworks generate realistic images through iterative denoising and have recently been adapted for brain-to-image reconstruction [19, 20]. Several studies have successfully integrated diffusion models with neural decoding to reconstruct natural images from fMRI signals, achieving significant improvements over earlier GAN- and VAE-based methods [15, 16, 21, 22, 27, 36]. Nevertheless, most existing approaches primarily emphasize semantic reconstruction, while structural preservation and cross-subject generalization remain challenging.

Multi-subject visual reconstruction is still a difficult research problem because each individual exhibits unique anatomical characteristics and distinct voxel activation patterns. These variations limit the ability of models to establish a common feature space across subjects [24, 33–35, 37]. Existing frameworks are largely optimized for individual subjects and often neglect complementary structural information and retrieval-based visual priors, limiting reconstruction fidelity and scalability across different participants.

To address these limitations, this work proposes a subject-agnostic multi-subject fMRI visual reconstruction framework that integrates shared latent representation learning, transformer-based neural encoding, multimodal CLIP-DINOv2 feature alignment, dual diffusion prior refinement, retrieval-guided matching, and Stable Diffusion-based image synthesis. The proposed framework jointly exploits semantic and structural visual representations while reducing inter-subject variability through a shared latent space, resulting in improved reconstruction fidelity and robust cross-subject generalization. Extensive experiments on the Haxby

and Natural Scenes Dataset (NSD) demonstrate the effectiveness of the proposed approach through both qualitative and quantitative evaluations.

The major contributions of this work are summarized as follows:

1. A subject-agnostic fMRI visual reconstruction framework is proposed using shared latent alignment and Transformer-based representation learning to reduce inter-subject variability.
2. A multimodal neural decoding strategy is developed by jointly predicting CLIP semantic and DINOv2 structural embeddings from latent fMRI representations.
3. A dual diffusion-prior and retrieval-guided Stable Diffusion framework is introduced to refine multimodal embeddings and improve reconstruction quality.
4. Extensive experiments on the Haxby and Natural Scenes Dataset (NSD) demonstrate improved reconstruction performance and semantic consistency compared with the evaluated baseline methods.

2 Related Work

Visual reconstruction from fMRI has advanced from statistical decoding to deep learning and generative models. Despite these advances, inter-subject Variability and fine-grained visual reconstruction remain challenging. Recent approaches addressing these issues are summarized in Table 1.

2.1 Classical Brain Decoding

Early fMRI brain decoding methods demonstrated that visual information could be recovered from neural activity using statistical encoding and decoding techniques, enabling the recognition of visual stimuli and object categories from cortical responses [3–5]. Conventional machine learning approaches, including support vector machines and Bayesian models, were subsequently employed to improve decoding performance [7, 28, 29]. However, these methods relied on hand-crafted features and linear modeling, limiting their ability to capture the complex nonlinear relationship between brain activity and visual representations.

2.2 Deep Learning-Based Brain Decoding

Deep neural network-based approaches improved neural feature extraction and decoding over conventional machine learning methods [9]. Subsequent work

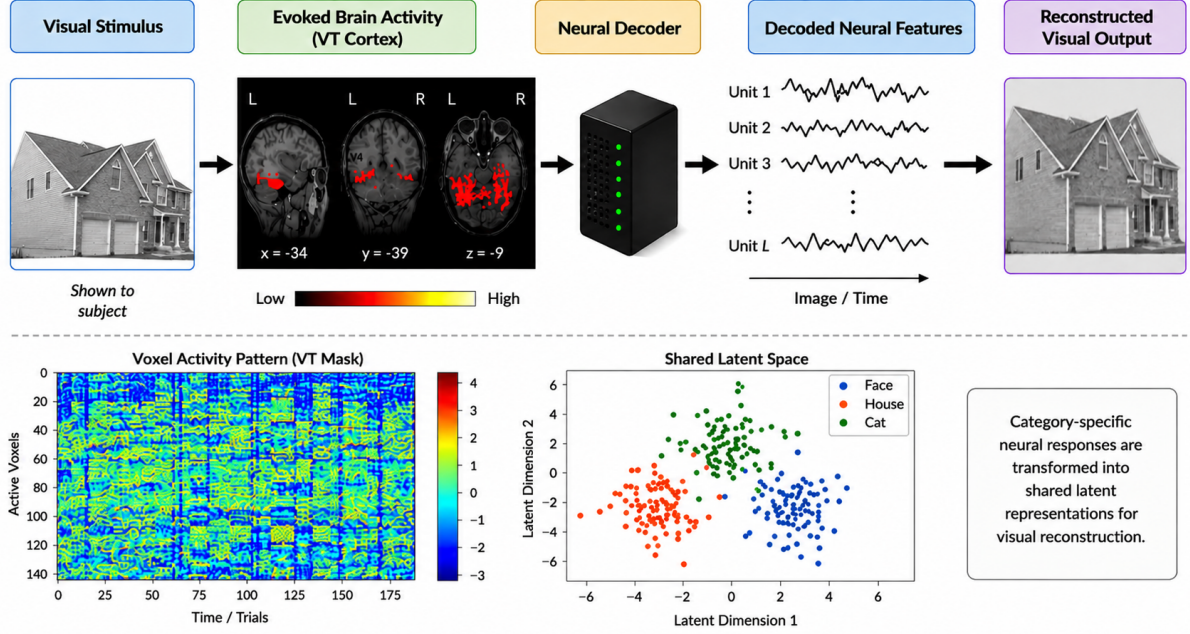


Figure 1: Overview of visual reconstruction from Haxby fMRI signals.

adopted nonlinear brain–image mappings and end-to-end models to enhance reconstruction quality [10–12]. More recently, Transformer-based methods have strengthened neural representation learning by capturing long-range dependencies through self-attention [13, 25]. However, achieving high-fidelity image reconstruction remains a challenge.

2.3 Multi-Subject Neural Alignment

Most fMRI decoding frameworks are developed for subject-specific learning, which limits their ability to generalize across individuals due to anatomical and functional variability in neural responses [24, 35]. To address this challenge, cross-subject alignment techniques, including shared response models, hyperalignment, and shared latent representation learning, have been proposed to transform subject-specific neural responses into a common representational space [2, 33, 34]. More recent studies have further improved multi-subject decoding through shared-subject learning and sparse voxel alignment strategies [24, 35, 37]. Despite these advances, achieving robust semantic and structural consistency across subjects remains a challenging problem for accurate visual reconstruction.

2.4 Diffusion-Based Brain Reconstruction

Diffusion models have substantially improved brain-to-image reconstruction by generating realistic images through iterative denoising processes [19, 20]. Recent studies have successfully integrated latent and stable diffusion models with fMRI decoding to reconstruct

visual stimuli with enhanced semantic accuracy and perceptual quality [16, 21, 22, 27]. Retrieval-guided and diffusion-prior frameworks have enhanced reconstruction performance through semantic visual priors [15, 24, 36]. However, preserving structural details and achieving robust cross-subject generalization remain challenging.

2.5 Multimodal Retrieval Decoding

The development of multimodal foundation models has substantially advanced neural decoding by enabling effective alignment of heterogeneous feature representations. CLIP provides a shared semantic embedding space for image–text representation learning, while DINO and DINOv2 capture rich structural and visual features [14, 17, 18]. Combining semantic and structural representations improves the preservation of both high-level object semantics and fine-grained visual details during image reconstruction. Although retrieval-guided and multi-subject reconstruction methods have enhanced reconstruction robustness, their joint integration remains an open research challenge [15, 24].

2.6 Research Gap

As summarized in Table 1, recent studies have improved fMRI-based visual reconstruction; however, several challenges remain. Existing methods often exhibit limited cross-subject generalization due to inter-subject variability and insufficient alignment of shared neural representations [15, 16, 21, 24]. Furthermore, transformer-based feature learning, semantic-structural

Table 1: Chronological evolution of diffusion-based studies for brain-to-image reconstruction.

Study (Year)	Task	Core Method	Dataset	Key Contribution
Ho et al. (2020) [19]	Image Generation	DDPM	Generic Images	Introduced denoising diffusion probabilistic models for high-quality image synthesis.
Rombach et al. (2022) [20]	Image Synthesis	Latent Diffusion	LAION/Image Datasets	Proposed latent diffusion for efficient high-resolution image generation.
Ozcelik and VanRullen (2023) [21]	Brain-to-Image Reconstruction	Generative Diffusion	Latent NSD	Applied latent diffusion for natural scene reconstruction from fMRI signals.
Takagi and Nishimoto (2023) [16]	Brain-to-Image Reconstruction	Stable Diffusion	NSD	First Stable Diffusion-based high-resolution image reconstruction from brain activity.
Meng and Yang (2023) [22]	Brain-to-Image Reconstruction	Dual-Guided Diffusion	Visual Stimulus fMRI	Improved reconstruction using dual guidance from neural features.
Chen et al. (2023) [27]	Brain-to-Image Reconstruction	Conditional Diffusion with Masked Guidance	NSD	Enhanced reconstruction using masked conditional diffusion guidance.
Ferrante et al. (2024) [15]	Retrieval-guided Reconstruction	Latent Diffusion + Retrieval	fMRI Dataset	Combined retrieval priors with latent diffusion for improved semantic reconstruction.
Scotti et al. (2024) [24]	Multi-subject Reconstruction	Shared-Subject Diffusion Decoding	NSD	Enabled robust shared-subject brain-to-image reconstruction with reduced subject-specific data.

feature fusion, and retrieval-guided diffusion have largely been explored independently rather than within a unified framework. Therefore, this work proposes a subject-agnostic reconstruction framework that integrates Transformer-based encoding, multimodal feature fusion, shared latent alignment, and retrieval-guided diffusion to improve cross-subject visual reconstruction.

3 Methodology

This section presents the proposed framework for subject-agnostic multi-subject fMRI visual reconstruction. The approach integrates shared latent alignment, Transformer-based neural representation learning, CLIP-DINOv2 semantic-structural feature prediction, dual diffusion-prior refinement, retrieval-guided matching, and Stable Diffusion-based image reconstruction. The complete processing pipeline is illustrated in Figure 2.

3.1 Problem Formulation

Let $\mathcal{X} = \{X_1, X_2, \dots, X_S\}$ denote the set of subject-specific fMRI measurements, where each $X_s \in R^{T \times V}$ represents the time-varying voxel responses of subject s , with T time points and V voxel features. The objective is to learn a subject-agnostic mapping from these

neural responses to a visual representation that can generate the corresponding image. After shared latent alignment, Transformer encoding, multimodal feature prediction, diffusion-prior refinement, and retrieval, the final image is reconstructed as

$$\hat{Y} = D(I_r, Z_f), \quad (1)$$

where Z_f is the fused semantic-structural representation, I_r is the retrieved reference image, $D(\cdot, \cdot)$ denotes the retrieval-guided Stable Diffusion model, and $\hat{Y}$ is the reconstructed image.

3.2 Overall Framework Overview

The proposed framework follows seven sequential stages: fMRI preprocessing, shared latent alignment, Transformer-based neural representation learning, CLIP-DINOv2 feature prediction, dual diffusion-prior refinement and multimodal fusion, retrieval-guided matching, and Stable Diffusion-based reconstruction. These stages progressively transform multi-subject fMRI responses into a unified visual representation and finally into a reconstructed image. The complete workflow is shown in Figure 2, and the procedure is summarized in Algorithm ??.

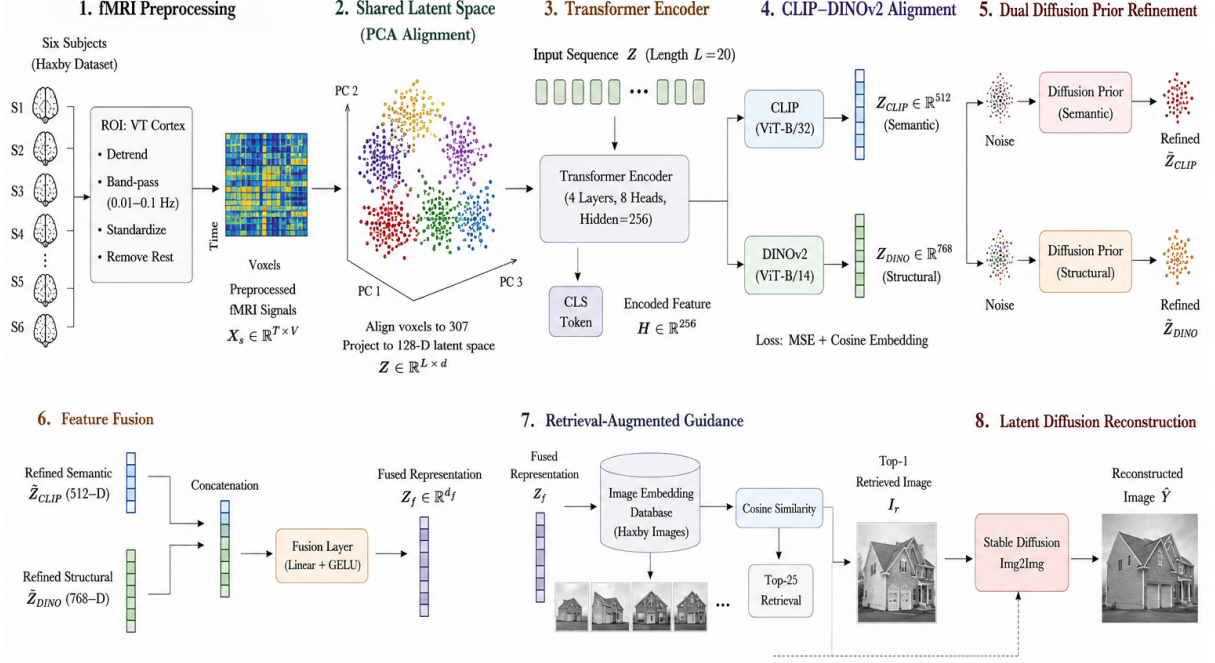


Figure 2: Overall architecture of the proposed subject-agnostic, multi-subject fMRI visual reconstruction framework.

3.3 Dataset Description

Three complementary fMRI datasets were used to evaluate the proposed framework: Haxby, the Natural Scenes Dataset (NSD), and the LIVE dataset collected at NIMHANS, Bengaluru. Haxby was used for category-level visual reconstruction and includes six subjects viewing seven visual categories: face, house, cat, shoe, chair, bottle, and scissors. Subjects S1–S4 were used for training, S5 for validation, and S6 for testing. The analysis used 307 aligned voxels from the ventral temporal (VT) cortex, with 12 runs per subject and a TR of 3.0 s.

NSD was used for natural-scene reconstruction using images associated with Microsoft COCO annotations. Four subjects, S1, S2, S5, and S7, were selected, with S1–S2 for training, S5 for validation, and S7 as the held-out test subject. The voxel dimensions were 15,724, 14,278, 13,039, and 12,682 for S1, S2, S5, and S7, respectively. The processed dataset contained 111,000 samples, comprising 55,500 training, 27,750 validation, and 27,750 test samples. NSD was acquired with a TR of 1.6 s and included multiple sessions with 12 runs per session. Data-availability settings of 1 h, 7 h, and 40 h were additionally evaluated using S1, S1–S2, and S1, S2, S5, and S7, respectively.

The LIVE dataset, collected at NIMHANS, Bengaluru, was used for independent cross-subject evaluation of natural-scene visual reconstruction. Six subjects were considered, with S1–S4 used for training, S5 for validation, and S6 for testing. Each subject provides a

4D BOLD volume of size $147 \times 144 \times 36 \times 165$, with 165 temporal samples and a TR of 3.0 s. The visual stimuli consist of natural-scene images linked to COCO image annotations. The LIVE data therefore provide an additional benchmark for assessing subject-wise generalization under a distinct fMRI acquisition setting.

3.4 fMRI Data Preprocessing

Raw fMRI BOLD signals contain temporal fluctuations and subject-dependent differences that can affect subsequent neural representation learning. The cross-subject activation maps in Figure 3 illustrate these differences across Haxby, NSD, and LIVE, with variations in the spatial distribution and intensity of brain activity. The mean BOLD responses shown in Figure 4 further demonstrate the temporal characteristics of the three datasets and motivate dataset-specific preprocessing.

For Haxby, rest and non-stimulus samples were removed, followed by VT ROI extraction, detrending, band-pass filtering, and voxel-wise standardization. NSD used the preprocessed MindEye HDF5 data with *StandardScaler*-based normalization. LIVE underwent VT ROI extraction, detrending, band-pass filtering, and voxel-wise standardization. The resulting representations were used for shared latent alignment.

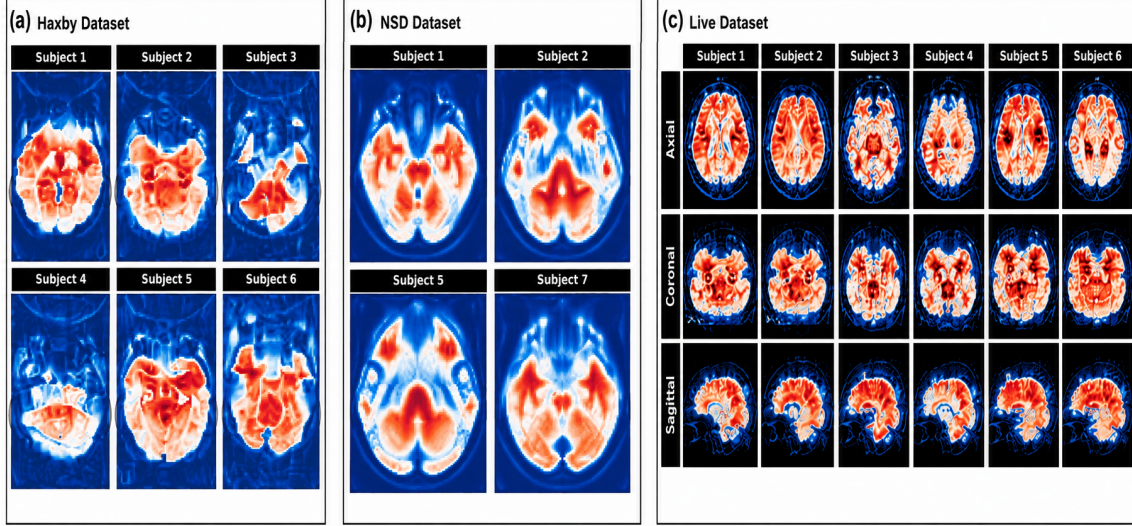


Figure 3: Cross-subject brain activation maps across three fMRI datasets: (a) Haxby, (b) Natural Scenes Dataset (NSD), and (c) LIVE. The maps illustrate subject-wise variability in the spatial distribution and intensity of brain activity during visual stimulation. For the LIVE dataset, a 3D visualization is additionally provided at the same time point to highlight the spatial heterogeneity of activation across subjects and to facilitate further exploration of its three-dimensional functional characteristics.

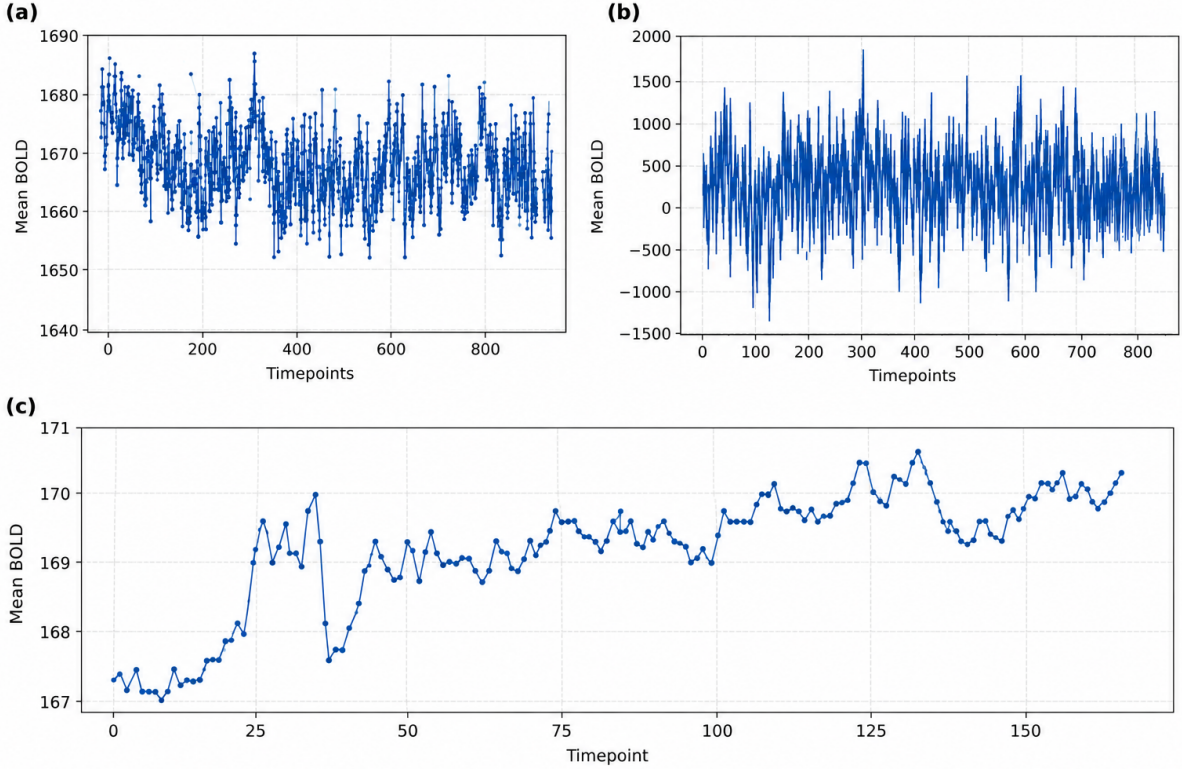


Figure 4: Mean BOLD signal responses across three benchmark datasets: (a) Haxby, (b) Natural Scenes Dataset (NSD), and (c) LIVE Dataset.

3.5 Shared Latent Representation Learning

After preprocessing, the fMRI responses are mapped to a shared latent space to reduce inter-subject variability.

As shown in Figure 5, subjects exhibit different voxel-wise activation patterns at corresponding time points despite performing the same visual task. These differences are also reflected in the BOLD responses. The

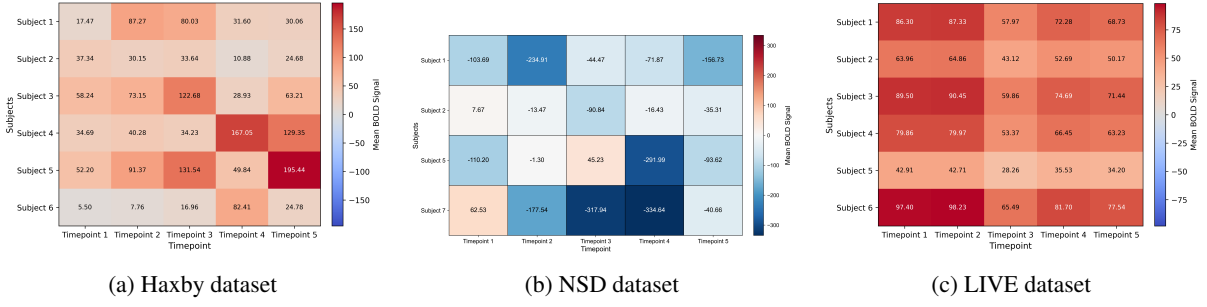


Figure 5: Cross-subject brain activity maps across three fMRI datasets: (a) Haxby, (b) Natural Scenes Dataset (NSD), and (c) LIVE. The maps illustrate subject-wise differences in the spatial distribution and intensity of brain activity.

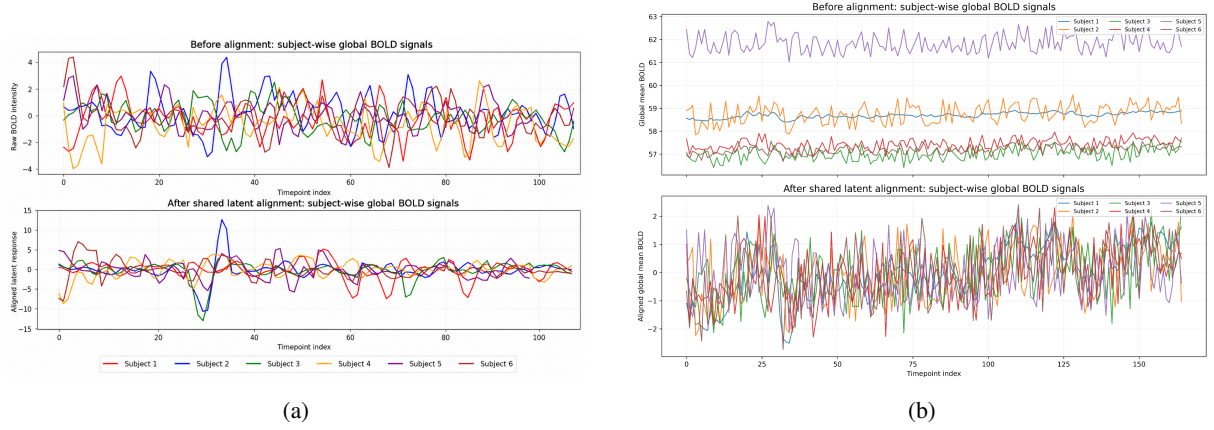


Figure 6: **Shared latent alignment across subjects.** (a) **Haxby dataset:** Alignment reduces subject-specific temporal and amplitude variability. (b) **LIVE dataset:** Alignment reduces inter-subject differences and produces a more consistent common latent representation.

proposed alignment transforms the subject-specific responses into a common representation,

$$Z = \Psi(X; \theta), \quad Z \in R^{N \times d}, \quad (2)$$

where $X \in R^{N \times V}$ denotes the preprocessed responses and d is the shared latent dimension. The alignment parameters are learned by minimizing $\mathcal{L}_{\text{align}}$. As illustrated in Figure 6, the aligned representations show greater consistency across subjects. The resulting shared features are then used by the Transformer for subsequent neural representation learning.

3.6 Transformer-Based Neural Representation Learning

The shared latent representation $Z \in R^{L \times d}$ is subsequently encoded using a Transformer encoder to learn subject-independent neural representations by modeling long-range dependencies among temporally ordered fMRI features. Here, $L = 20$ denotes the sequence length and $d = 128$ is the latent feature dimension. An input projection first maps the latent features into a 256-dimensional embedding space, followed by the ad-

dition of a learnable CLS token and positional encoding to preserve sequence order. Multi-head self-attention computes contextual relationships as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (3)$$

where Q , K , and V denote the query, key, and value matrices, respectively, and d_k is the key dimension. The encoder consists of four Transformer layers with eight attention heads, feed-forward networks, residual connections, layer normalization, and dropout ($p = 0.1$). The resulting global neural representation is obtained from the CLS token,

$$H = T(Z), \quad H \in R^{256}, \quad (4)$$

where $T(\cdot)$ denotes the Transformer encoder. The learned latent feature H is subsequently projected into the CLIP semantic and DINOv2 structural embedding branches for multimodal representation learning.

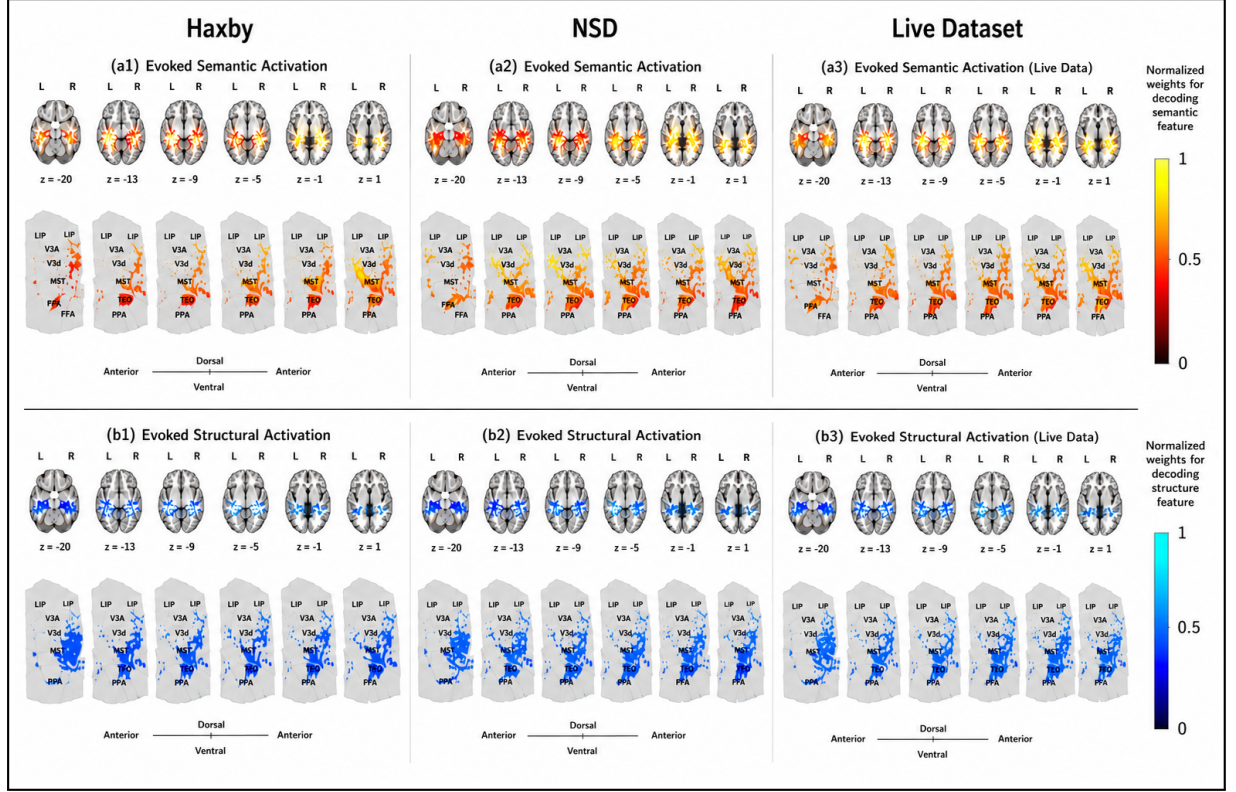


Figure 7: Comparison of semantic and structural voxel activation patterns in the Haxby and Natural Scenes Dataset (NSD), illustrating representative distributions of informative voxels used for visual decoding.

3.7 CLIP-DINOv2 Semantic-Structural Feature Alignment

The shared latent representation $H \in R^{256}$ is projected into semantic and structural spaces to capture complementary information from brain activity. As illustrated in Figure 7, the semantic representation encodes high-level visual meaning, while the structural representation captures fine-grained spatial patterns in the corresponding neural activity. The embedding functions are defined as

$$Z_{\text{CLIP}} = E_{\text{CLIP}}(H), \quad Z_{\text{DINO}} = E_{\text{DINO}}(H), \quad (5)$$

where $E_{\text{CLIP}}(\cdot)$ and $E_{\text{DINO}}(\cdot)$ denote the CLIP and DINOv2 projection networks, producing semantic embeddings $Z_{\text{CLIP}} \in R^{512}$ and structural embeddings $Z_{\text{DINO}} \in R^{768}$, respectively. The projection networks are optimized using a joint embedding objective

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda \mathcal{L}_{\text{Cos}}, \quad (6)$$

where $\mathcal{L}_{\text{MSE}}$ preserves feature reconstruction accuracy, $\mathcal{L}_{\text{Cos}}$ maximizes embedding similarity, and λ balances the two objectives. The resulting semantic and structural embeddings are subsequently refined by the dual diffusion prior.

3.8 Dual Diffusion Prior Refinement

To refine the predicted semantic and structural embeddings, separate diffusion priors are applied to the CLIP and DINOv2 representations. The refinement process is formulated as

$$\tilde{Z}_{\text{CLIP}} = P_{\text{CLIP}}(Z_{\text{CLIP}}), \quad \tilde{Z}_{\text{DINO}} = P_{\text{DINO}}(Z_{\text{DINO}}), \quad (7)$$

where $P_{\text{CLIP}}(\cdot)$ and $P_{\text{DINO}}(\cdot)$ denote the semantic and structural diffusion prior networks, respectively. The refined embeddings are subsequently fused into a unified latent representation

$$Z_f = \phi(\tilde{Z}_{\text{CLIP}}, \tilde{Z}_{\text{DINO}}), \quad (8)$$

where $\phi(\cdot)$ denotes the feature fusion function. The priors are optimized using

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda \mathcal{L}_{\text{Cos}} + \gamma \|W\|_2^2, \quad (9)$$

where $\mathcal{L}_{\text{MSE}}$ the minimized reconstruction error between the predicted and target embeddings, $\mathcal{L}_{\text{Cos}}$ preserves embedding similarity, $\|W\|_2^2$ is the weight regularization term, and λ and γ are balancing coefficients. The resulting fused representation Z_f is subsequently used for retrieval-guided diffusion-based image reconstruction.

3.9 Retrieval-Augmented Guidance and Latent Diffusion Reconstruction

The fused latent representation Z_f is used to retrieve visually similar reference images from the retrieval database. Given the fused embedding and database feature E_i , the similarity score is computed as

$$S_i = \frac{Z_f^T E_i}{\|Z_f\| \|E_i\|}. \quad (10)$$

The retrieved reference image is selected as

$$I_r = \arg \max_{I_i} S_i, \quad (11)$$

where the Top- K nearest neighbours are searched and the highest-ranked image is used for reconstruction. The retrieved image and fused latent representation jointly condition the pretrained latent diffusion model,

$$\hat{Y} = D(I_r, Z_f), \quad (12)$$

where $D(\cdot)$ denotes the pretrained Stable Diffusion Img2Img model and $\hat{Y}$ is the reconstructed image. The reconstruction performance is quantitatively evaluated using a comprehensive set of objective metrics that assess the fidelity, perceptual quality, semantic consistency, and retrieval effectiveness of the reconstructed images.

3.10 Training Strategy and Evaluation Metrics

The proposed framework was implemented in PyTorch and trained on paired multi-subject fMRI data $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^N$ using mixed-precision computation on an NVIDIA RTX A6000 GPU. Each normalized fMRI sequence $X_i \in \mathbb{R}^{20 \times 128}$ was encoded by a four-layer Transformer with eight attention heads, a feed-forward dimension of 512, GELU activation, positional encoding, and a learnable CLS token to obtain the latent representation $H \in \mathbb{R}^{256}$ [25]. The latent features were projected to semantic and structural spaces through CLIP ($256 \rightarrow 1024 \rightarrow 1024 \rightarrow 512$) and DINOv2 ($256 \rightarrow 1024 \rightarrow 768$) projectors, refined by dual diffusion priors, and fused for retrieval-guided Stable Diffusion reconstruction [14, 18, 20, 22]. The network was optimized using a joint reconstruction objective with AdamW in three stages: Transformer training (15 epochs, batch size 32), CLIP-DINOv2 alignment (40 epochs, batch size 16), and diffusion prior refinement (20 epochs, batch size 16) [19, 25].

Reconstruction performance was evaluated at pixel, feature, and semantic levels. Pixel-level fidelity was assessed using SSIM, AlexNet, and PixCorr. Feature-level similarity was measured using pretrained Incep-

tionV3, EfficientNet-B0, and SwAV embeddings. Semantic consistency between reconstructed and ground-truth images was quantified using OpenCLIP cosine similarity [14]. To validate performance improvements across the seven visual categories, statistical analysis was performed using paired Student's t -test and Wilcoxon signed-rank test. Cohen's d and 95% confidence intervals were further computed to measure effect size and result reliability, with statistical significance defined at $p < 0.05$.

4 Results and Discussion

This section evaluates the proposed multi-subject fMRI visual reconstruction framework on the Haxby, NSD, and LIVE datasets. Model convergence is first analyzed using the training and validation losses, followed by quantitative and qualitative evaluation of the reconstructed images. Figure 8 shows stable convergence across the three datasets, providing a basis for assessing the reconstruction performance.

4.1 Quantitative Reconstruction Performance

The proposed alignment achieves consistent cross-subject reconstruction compared with existing approaches [33–35, 37]. On Haxby, the mean SSIM and PixCorr reach 0.8184 and 0.6086, respectively. NSD yields 0.5333 SSIM, 0.4304 PixCorr, and 95.79% CLIP similarity, indicating effective preservation of visual information.

The comparative results in Figure 9 further demonstrate the advantage of the proposed framework over existing methods. Compared with MindEye2, the proposed method improves SSIM and PixCorr by approximately 23.7% and 33.5%, respectively, while CLIP similarity increases by about 3.0%. The EffNet-B and SwAV distances are also reduced by approximately 19.2% and 20.3%, indicating improved perceptual feature agreement. These results support the benefit of combining shared latent alignment with semantic-structural diffusion refinement.

The data-efficiency analysis in Table 6 shows a consistent improvement with increasing training data. From 1 h to 40 h, SSIM increases by approximately 16.8% and PixCorr by 22.5%, while CLIP similarity improves progressively and both feature distances decrease. Finally, the LIVE results in Table 5 maintain consistent subject-wise performance, with average SSIM of 0.3917 and CLIP similarity of 74.16%, demonstrating that the proposed shared representation remains applicable across different fMRI settings.

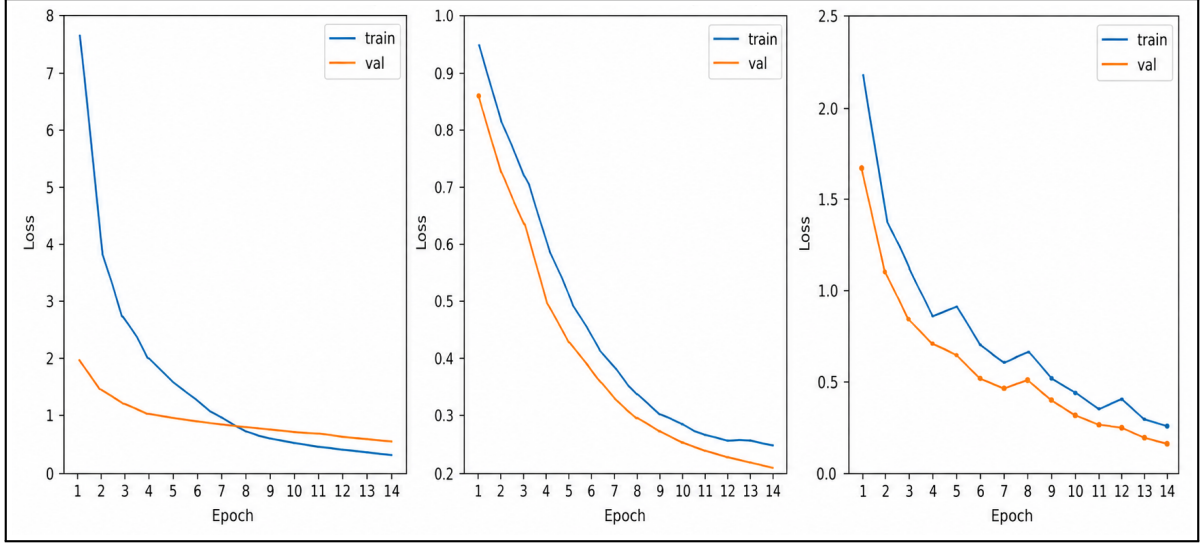


Figure 8: Training and validation loss curves of the proposed framework on the (a) Haxby , (b) NSD and c) live datasets, demonstrating stable convergence during model optimization.

Table 2: Category-wise quantitative reconstruction performance on the Haxby dataset.

Category	Low-Level					High-Level		
	SSIM $\uparrow$	PixCorr $\uparrow$	AlexNet(2) $\uparrow$	AlexNet(5) $\uparrow$	Inception $\uparrow$	CLIP $\uparrow$	EffNet-B $\downarrow$	SwAV $\downarrow$
Scissors	0.8406	0.2391	98.52%	99.12%	59.89%	84.61%	0.5457	0.8526
Shoe	0.9249	0.9846	99.91%	99.82%	94.32%	87.32%	0.9111	0.9485
House	0.8665	0.9711	99.87%	99.94%	95.02%	88.70%	0.5387	0.7108
Cat	0.9095	0.9921	99.91%	99.86%	91.85%	91.10%	0.8766	0.9240
Bottle	0.7866	0.0827	97.23%	98.68%	62.91%	65.66%	0.4652	0.7968
Face	0.7840	0.7941	99.84%	99.76%	89.24%	97.02%	0.9212	0.9443
Chair	0.6170	0.1964	94.49%	93.39%	19.39%	55.36%	0.3424	0.7134

Table 3: Subject-wise quantitative reconstruction performance on the Haxby dataset.

Subject	Low-Level					High-Level		
	SSIM $\uparrow$	PixCorr $\uparrow$	AlexNet(2) $\uparrow$	AlexNet(5) $\uparrow$	Inception $\uparrow$	CLIP $\uparrow$	EffNet-B $\downarrow$	SwAV $\downarrow$
Subject 1	0.8012	0.5821	98.24%	98.46%	71.18%	80.15%	0.6912	0.8623
Subject 2	0.8128	0.5984	98.43%	98.55%	72.46%	80.84%	0.7096	0.8731
Subject 3	0.8216	0.6148	98.62%	98.71%	73.65%	81.57%	0.7195	0.8794
Subject 4	0.8279	0.6235	98.73%	98.86%	74.41%	82.08%	0.7268	0.8857
Subject 5	0.8185	0.6102	98.56%	98.69%	73.27%	81.43%	0.7171	0.8786
Subject 6	0.8284	0.6227	98.68%	98.63%	74.41%	82.27%	0.7479	0.8919
Average	0.8184	0.6086	98.54%	98.65%	73.23%	81.39%	0.7187	0.8785

Table 4: Subject-wise quantitative reconstruction performance on the NSD dataset.

Subject	Low-Level					High-Level		
	SSIM $\uparrow$	PixCorr $\uparrow$	AlexNet(2) $\uparrow$	AlexNet(5) $\uparrow$	Inception $\uparrow$	CLIP $\uparrow$	EffNet-B $\downarrow$	SwAV $\downarrow$
Subject 1	0.5247	0.4213	97.84%	99.23%	97.36%	95.47%	0.5128	0.2816
Subject 2	0.5314	0.4286	98.07%	99.34%	97.52%	95.73%	0.5034	0.2769
Subject 5	0.5392	0.4361	98.26%	99.57%	97.83%	96.04%	0.4957	0.2713
Subject 7	0.5385	0.4354	98.13%	99.46%	97.71%	95.92%	0.4896	0.2685
Average	0.5333	0.4304	98.08%	99.40%	97.61%	95.79%	0.5004	0.2746

4.2 Qualitative Visual Reconstruction Results

The qualitative results in Figures 10a, 10b, and 10c visually support the quantitative results on the Haxby,

Table 5: Subject-wise quantitative reconstruction performance on the Live fMRI dataset.

Subject	Low-Level				High-Level			
	SSIM $\uparrow$	PixCorr $\uparrow$	AlexNet(2) $\uparrow$	AlexNet(5) $\uparrow$	Inception $\uparrow$	CLIP $\uparrow$	EffNet-B $\downarrow$	SwAV $\downarrow$
Subject 1	0.3847	0.3618	76.42%	81.35%	69.84%	72.61%	0.6842	0.6127
Subject 2	0.3926	0.3745	78.16%	83.27%	71.52%	74.38%	0.6618	0.5943
Subject 3	0.4078	0.3896	80.43%	85.16%	74.28%	76.15%	0.6387	0.5716
Subject 4	0.3769	0.3497	73.84%	79.42%	67.91%	70.84%	0.7125	0.6384
Subject 5	0.4015	0.3827	79.62%	84.31%	75.16%	77.43%	0.6214	0.5589
Subject 6	0.3868	0.3659	76.91%	82.08%	70.42%	73.57%	0.6736	0.6031
Average	0.3917	0.3707	77.56%	82.60%	71.52%	74.16%	0.6654	0.5965

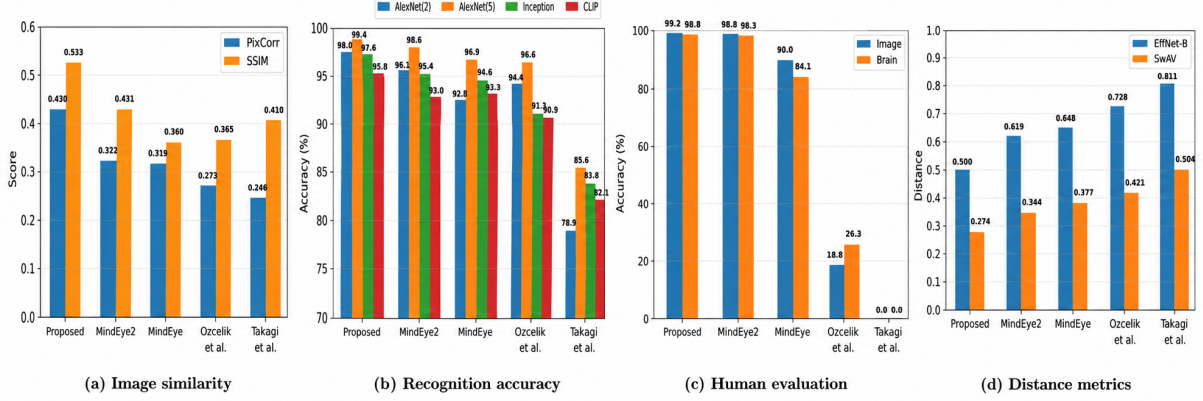


Figure 9: Quantitative performance comparison of the proposed framework and existing methods on the NSD dataset.

Table 6: Data-efficiency analysis of the proposed framework on the NSD dataset using different amounts of training data while keeping the validation and test subjects fixed.

Training Data	Low-Level Metrics				High-Level Metrics			
	SSIM $\uparrow$	PixCorr $\uparrow$	AlexNet(2) $\uparrow$	AlexNet(5) $\uparrow$	CLIP $\uparrow$	Inception $\uparrow$	EffNet-B $\downarrow$	SwAV $\downarrow$
1 hr	0.4567	0.3513	95.94%	98.37%	93.28%	95.06%	0.6428	0.3516
7 hr	0.4984	0.3926	97.23%	98.94%	94.76%	96.43%	0.5617	0.3074
40 hr	0.5333	0.4304	98.08%	99.40%	95.79%	97.61%	0.5004	0.2746

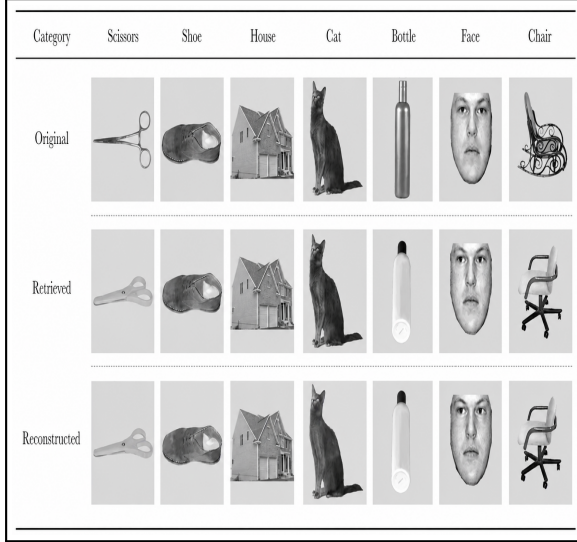
NSD, and LIVE datasets. The CLIP–DINOv2-guided retrieval provides semantically and structurally relevant reference images, which help the diffusion model generate reconstructions that preserve the main objects, scene structure, and visual content. This is consistent with the reported Haxby SSIM of 0.8184 and PixCorr of 0.6086, NSD SSIM of 0.5333 and CLIP similarity of 95.79%, and LIVE SSIM of 0.3917 and CLIP similarity of 74.16%.

Figure 11a demonstrates subject-wise variation on the NSD dataset, where the same stimulus produces visually different reconstructions across subjects. Despite these appearance differences caused by inter-subject neural variability, the reconstructed images retain the common semantic identity and major visual structure of the stimulus. This observation agrees with the relatively consistent subject-wise quantitative performance, particularly the NSD CLIP similarity ranging from 95.47% to 96.04%.

Figure 11b compares the proposed method with Ozcelik et al., MindEye, MindEye2, and other existing approaches. The proposed method produces reconstructions with stronger semantic correspondence and better preservation of structural details than the compared methods. These visual improvements are consistent with the quantitative comparison in Figure 9, where the proposed framework achieves improved SSIM, PixCorr, and CLIP similarity together with lower EffNet-B and SwAV distances, demonstrating the effectiveness of shared latent alignment and CLIP–DINOv2-guided diffusion reconstruction.

4.3 Ablation Study

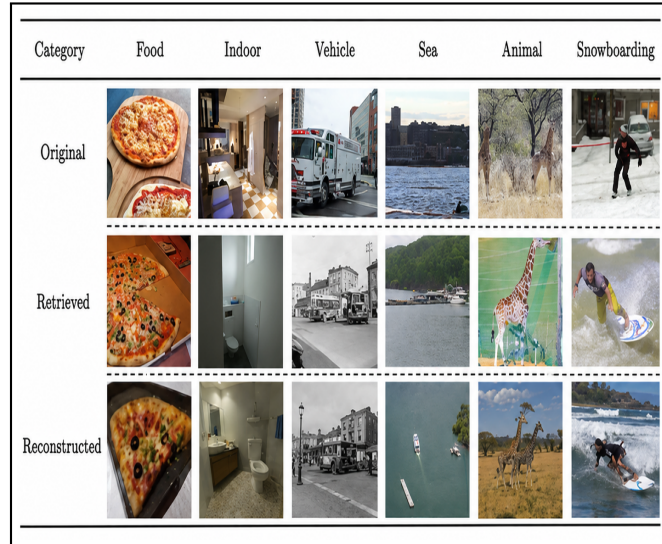
The ablation study evaluates the progressive contribution of semantic guidance, retrieval, and structural representation across the Haxby, NSD, and LIVE datasets. As shown in Table 7, reconstruction quality improves



(a) Qualitative reconstruction results on the Haxby dataset.



(b) Qualitative reconstruction results on the NSD dataset.



(c) Qualitative reconstruction results on the LIVE dataset.

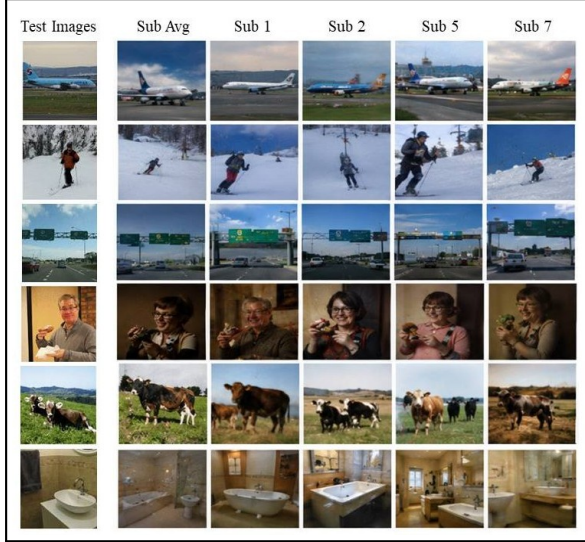
Figure 10: **Qualitative reconstruction results of the proposed framework across three datasets.** The results show the original stimulus, retrieved reference image, and reconstructed output for the Haxby, NSD, and LIVE datasets, demonstrating the generalization capability of the proposed framework across different fMRI datasets.

Table 7: Three-stage ablation results on the Haxby, NSD, and LIVE datasets.

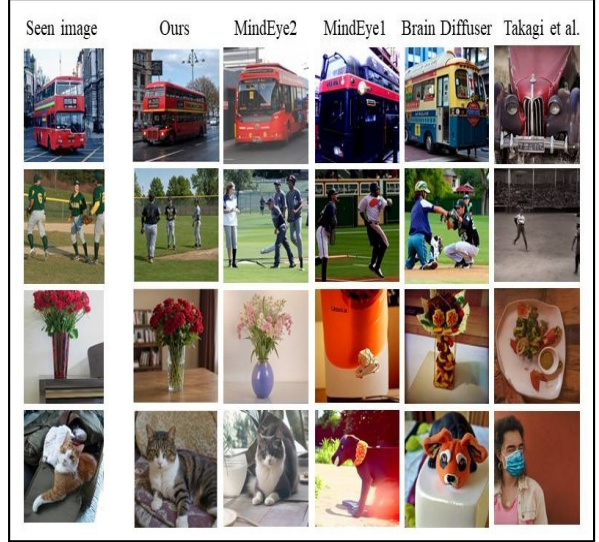
Dataset	Method	PixCorr $\uparrow$	SSIM $\uparrow$	AlexNet(2) $\uparrow$	AlexNet(5) $\uparrow$	Inception $\uparrow$	CLIP $\uparrow$	EffNet-B $\downarrow$	SwAV $\downarrow$
Haxby	CLIP Only	-0.0140	0.3661	91.28%	95.12%	41.91%	61.61%	0.8012	0.9156
	CLIP + Retrieval	0.2985	0.6228	96.14%	96.97%	51.46%	67.70%	0.7894	0.8421
	CLIP + DINOv2 + Retrieval	0.6086	0.8184	98.54%	98.65%	73.23%	81.39%	0.7187	0.8785
NSD	CLIP Only	0.3080	0.3960	94.30%	97.20%	92.80%	91.60%	0.6640	0.3890
	CLIP + Retrieval	0.3900	0.4490	95.80%	98.10%	94.80%	92.90%	0.6240	0.3510
	CLIP + DINOv2 + Retrieval	0.4300	0.5333	98.08%	99.40%	97.61%	95.79%	0.5004	0.2746
LIVE	CLIP Only	0.2815	0.3027	70.84%	75.92%	64.37%	67.42%	0.7896	0.7215
	CLIP + Retrieval	0.3346	0.3568	74.28%	79.41%	68.95%	71.38%	0.7247	0.6418
	CLIP + DINOv2 + Retrieval	0.3707	0.3917	77.56%	82.60%	71.52%	74.16%	0.6654	0.5965

from CLIP Only to CLIP + Retrieval and further to the complete CLIP + DINOv2 + Retrieval configuration. Across all three datasets, the complete configuration obtains SSIM values of 0.8184 on Haxby, 0.5333 on NSD,

and 0.3917 on LIVE. The reconstructions presented in Figure 12 further demonstrate their ability to retain the principal visual content and structural organization of the input stimuli.



(a) Subject-wise generalization results on the Natural Scenes Dataset (NSD). The first column shows the ground-truth test images, followed by the shared latent representation (Subject Average) and reconstructions generated from Subjects 1, 2, 5, and 7.



(b) Qualitative comparison of the proposed framework with existing methods. The proposed model preserves semantic content, object structure, and visual details more effectively than previous approaches.

Figure 11: Qualitative evaluation of the proposed framework on the Natural Scenes Dataset (NSD). (a) Subject-wise generalization across multiple subjects using the learned shared latent representation. (b) Visual comparison with existing state-of-the-art reconstruction methods.

Table 8: Component-wise ablation of the proposed framework using SSIM.

Method	Haxby SSIM $\uparrow$	NSD SSIM $\uparrow$	LIVE SSIM $\uparrow$
MLP Baseline	0.5081	0.3324	0.2718
MLP + Shared Latent Alignment	0.5523	0.3617	0.2896
Transformer	0.5519	0.3695	0.2964
Transformer + Shared Latent Alignment	0.6255	0.3978	0.3149
CLIP Only	0.3661	0.3960	0.3027
DINOv2 Only	0.4218	0.4087	0.3186
CLIP + DINOv2	0.4816	0.4235	0.3327
CLIP + Retrieval	0.6228	0.4490	0.3568
CLIP + DINOv2 + Retrieval	0.7015	0.4816	0.3719
Full Model + Diffusion Prior	0.7722	0.5127	0.3846
Proposed Full Model	0.8184	0.5333	0.3917

Table 9: Top- K retrieval accuracy across the Haxby, NSD, and LIVE datasets.

Dataset	Top-1	Top-5	Top-10	Top-25	Best
Haxby	90.05%	89.04%	88.43%	87.42%	90.05%
NSD	91.24%	92.18%	89.56%	88.91%	89.56%
LIVE	79.31%	81.04%	80.26%	82.15%	82.15%

The component-wise analysis in Table 8 further demonstrates the contribution of shared latent alignment, Transformer representation, CLIP, DINOv2, retrieval, and diffusion refinement. The complete model

consistently provides the strongest SSIM across the three datasets. The retrieval analysis in Table 9 also shows high Top- K retrieval accuracy, with the best observed performance reaching 90.05% for Haxby,

Stimulus	Scissors	Shoe	House	Cat	Bottle	Face	Chair
Original Images							
CLIP							
CLIP + Retrieval							
Proposed Method							

(a) Haxby dataset.

Stimulus	Bus	Food	Horses	Surfing	Train	Street
Original Images						
CLIP						
CLIP + Retrieval						
Proposed Method						

(b) NSD dataset.

Stimulus	Food	Animal	Outdoor	Vehicle	Room	Indoor
Original Images						
CLIP						
CLIP + Retrieval						
Proposed Method						

(c) LIVE dataset.

Figure 12: Qualitative ablation results across Haxby, NSD, and LIVE datasets.

89.56% for NSD, and 82.15% for LIVE at their respective optimal retrieval levels. Overall, the improvements are statistically supported across the evaluated datasets, with significant results observed at the reported testing threshold ($p \leq 0.05$); the strongest reported effects are obtained for Haxby ($p = 0.0022$, Cohen’s $d = 1.31$), NSD ($p = 0.001$, Cohen’s $d = 1.39$), and LIVE ($p = 0.05$, Cohen’s $d = 1.19$). These results demonstrate the effectiveness of combining shared latent alignment, semantic–structural guidance, retrieval, and diffusion refinement.

5 Discussion

The proposed framework performs cross-subject fMRI visual reconstruction using the Haxby and NSD benchmark datasets and LIVE data collected at NIMHANS, Bengaluru, with subject-wise training, validation, and testing. Unlike earlier semantic- and retrieval-oriented methods such as Takagi et al., Ozcelik et al., MindEye, and MindEye2, the proposed approach combines **CLIP semantic features with DINOv2 structural features** in a shared latent space. The fused representation guides retrieval of relevant reference images, followed by latent diffusion refinement. The framework achieves SSIM values of **0.8184**, **0.5333**, and **0.3917** for Haxby, NSD, and LIVE, respectively, with **95.79% CLIP similarity on NSD**, while subject-wise experiments demon-

strate preservation of common stimulus information despite neural variability.

The quantitative and qualitative evaluations include subject-wise reconstruction, unseen-subject testing, average representations, data-efficiency experiments using **1 h, 7 h, and 40 h**, and comparison with existing methods on NSD. The visual comparisons indicate a clear enhancement in reconstruction quality as the framework progresses from CLIP-based decoding to retrieval-assisted and CLIP-DINOv2-guided reconstruction. The complete configuration delivers the highest performance across the evaluated datasets, while the retrieval module achieves peak accuracies of **90.05% for Haxby, 89.56% for NSD, and 82.15% for LIVE**, demonstrating effective selection of relevant visual references. Overall, the findings demonstrate that combining shared latent alignment, semantic-structural fusion, retrieval guidance, and latent diffusion improves cross-subject reconstruction beyond semantic guidance alone.

6 Conclusion

This study presents a unified framework for multi-subject fMRI-based visual reconstruction that addresses inter-subject variability through shared latent representation learning. By integrating Transformer-based neural encoding, CLIP-DINOv2 semantic-structural fusion, retrieval-guided conditioning, and latent diffusion, the proposed framework enhances both semantic correspondence and structural fidelity. Experiments on the Haxby, NSD, and LIVE datasets demonstrate consistent reconstruction performance across subjects and imaging settings, supported by quantitative, qualitative, and ablation analyses. Overall, the results demonstrate that combining semantic and structural representations within a shared latent space provides an effective approach for cross-subject brain-to-image reconstruction.

6.1 Future Work

Future work will focus on improving cross-subject alignment through advanced BOLD preprocessing and sparse neural representation learning to extract more informative visual features from fMRI signals. The framework will be extended to larger and more diverse neuroimaging datasets and to **3D visual reconstruction** for recovering richer spatial and volumetric information. Advanced diffusion models and multimodal semantic-structural fusion will also be explored to further improve reconstruction quality. In addition, explainable AI techniques will be incorporated to identify brain regions and neural patterns contributing to visual reconstruction, enabling more interpretable and reliable brain decoding.

Acknowledgment

The authors Prof. Ramesh K., Dr. Sheetalrani R. Kawale, and Pavitra Muttur gratefully acknowledge the financial support provided by the Vision Group on Science and Technology (VGST), Govt. of Karnataka, India, through the funded project under the GRE scheme (GRD No. 1115).

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